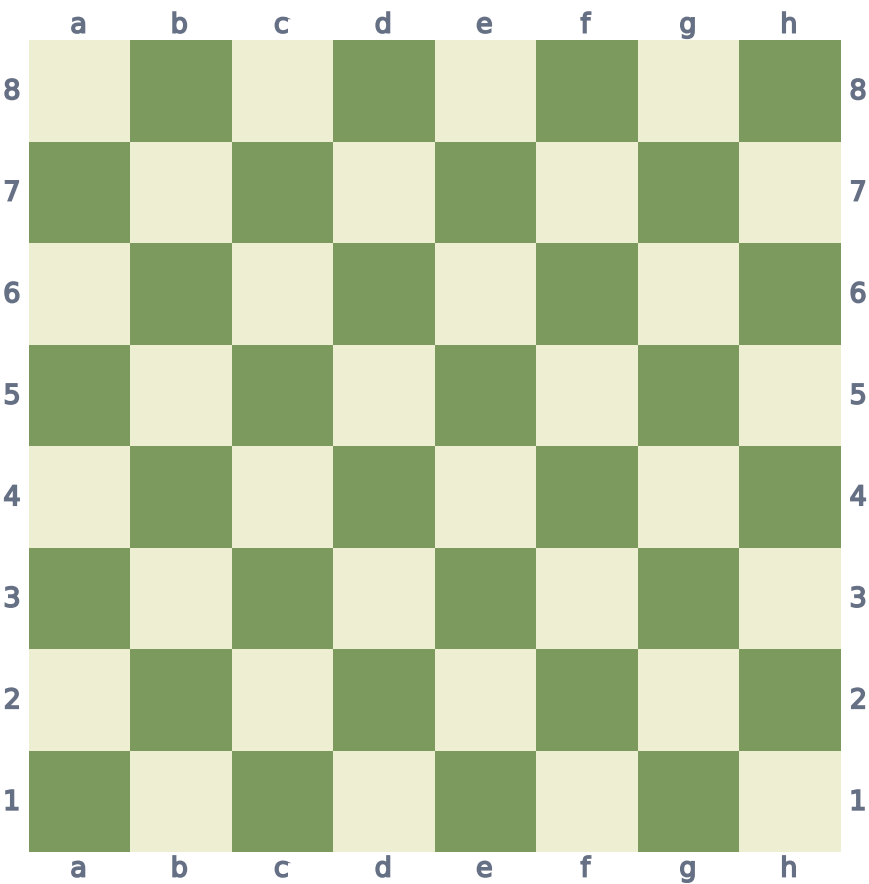


Visualization and Board-Vision Training

Coordinates, routes, rays, reconstruction, and blind calculation



Strengthen the internal board through progressive drills in square color, piece routes, long-range geometry, notation, and position reconstruction.

60 practice units | separated solutions | scoring | review plan

ORIENTATION

How to get real improvement from this course

PURPOSE

Strengthen the internal board through progressive drills in square color, piece routes, long-range geometry, notation, and position reconstruction.

TRAINING RULES

- Answer without looking at a board; use the diagram only during correction.
- Accuracy comes before speed. Shorten the time limit only after two perfect attempts.
- Say each square and piece location aloud to strengthen notation-to-image conversion.
- Stop when the internal board becomes unreliable; reset rather than practicing confusion.
-

COURSE SIZE

This PDF contains 60 practice units. Problems appear before their solutions whenever practical. Review misses after 1, 3, 7, 14, and 30 days.

VISUALIZATION DRILLS

Exercises 1-3

01. SQUARE COLOR

Without a board: is a1 light or dark? Name its four orthogonal neighbors that remain on the board.

02. SQUARE COLOR

Without a board: is b2 light or dark? Name its four orthogonal neighbors that remain on the board.

03. SQUARE COLOR

Without a board: is c3 light or dark? Name its four orthogonal neighbors that remain on the board.

VISUALIZATION DRILLS

Exercises 4-6

04. SQUARE COLOR

Without a board: is d4 light or dark? Name its four orthogonal neighbors that remain on the board.

05. SQUARE COLOR

Without a board: is e5 light or dark? Name its four orthogonal neighbors that remain on the board.

06. SQUARE COLOR

Without a board: is f6 light or dark? Name its four orthogonal neighbors that remain on the board.

VISUALIZATION DRILLS

Exercises 7-9

07. SQUARE COLOR

Without a board: is g7 light or dark? Name its four orthogonal neighbors that remain on the board.

08. SQUARE COLOR

Without a board: is h8 light or dark? Name its four orthogonal neighbors that remain on the board.

09. SQUARE COLOR

Without a board: is a8 light or dark? Name its four orthogonal neighbors that remain on the board.

VISUALIZATION DRILLS

Exercises 10-12

10. SQUARE COLOR

Without a board: is h1 light or dark? Name its four orthogonal neighbors that remain on the board.

11. SQUARE COLOR

Without a board: is c6 light or dark? Name its four orthogonal neighbors that remain on the board.

12. SQUARE COLOR

Without a board: is f3 light or dark? Name its four orthogonal neighbors that remain on the board.

VISUALIZATION DRILLS

Exercises 13-15

13. KNIGHT ROUTE

What is the minimum number of knight moves from a1 to h8? Visualize one legal route.

14. KNIGHT ROUTE

What is the minimum number of knight moves from b1 to e5? Visualize one legal route.

15. KNIGHT ROUTE

What is the minimum number of knight moves from g1 to e4? Visualize one legal route.

VISUALIZATION DRILLS

Exercises 16-18

16. KNIGHT ROUTE

What is the minimum number of knight moves from c3 to f7? Visualize one legal route.

17. KNIGHT ROUTE

What is the minimum number of knight moves from h1 to a8? Visualize one legal route.

18. KNIGHT ROUTE

What is the minimum number of knight moves from d4 to a1? Visualize one legal route.

VISUALIZATION DRILLS

Exercises 19-21

19. KNIGHT ROUTE

What is the minimum number of knight moves from e5 to h1? Visualize one legal route.

20. KNIGHT ROUTE

What is the minimum number of knight moves from a4 to h5? Visualize one legal route.

21. KNIGHT ROUTE

What is the minimum number of knight moves from b7 to g2? Visualize one legal route.

VISUALIZATION DRILLS

Exercises 22-24

22. KNIGHT ROUTE

What is the minimum number of knight moves from f3 to c7? Visualize one legal route.

23. KNIGHT ROUTE

What is the minimum number of knight moves from h6 to f1? Visualize one legal route.

24. KNIGHT ROUTE

What is the minimum number of knight moves from c1 to h6? Visualize one legal route.

VISUALIZATION DRILLS

Exercises 25-27

25. BISHOP RAYS

Name every square a bishop on d4 attacks on an empty board.

26. ROOK RAYS

Name every square a rook on d4 attacks on an empty board.

27. BISHOP RAYS

Name every square a bishop on e4 attacks on an empty board.

VISUALIZATION DRILLS

Exercises 28-30

28. ROOK RAYS

Name every square a rook on e4 attacks on an empty board.

29. BISHOP RAYS

Name every square a bishop on c3 attacks on an empty board.

30. ROOK RAYS

Name every square a rook on c3 attacks on an empty board.

VISUALIZATION DRILLS

Exercises 31-33

31. BISHOP RAYS

Name every square a bishop on f6 attacks on an empty board.

32. ROOK RAYS

Name every square a rook on f6 attacks on an empty board.

33. BISHOP RAYS

Name every square a bishop on a1 attacks on an empty board.

VISUALIZATION DRILLS

Exercises 34-36

34. ROOK RAYS

Name every square a rook on a1 attacks on an empty board.

35. BISHOP RAYS

Name every square a bishop on h8 attacks on an empty board.

36. ROOK RAYS

Name every square a rook on h8 attacks on an empty board.

VISUALIZATION DRILLS

Exercises 37-39

37. BISHOP RAYS

Name every square a bishop on b5 attacks on an empty board.

38. ROOK RAYS

Name every square a rook on b5 attacks on an empty board.

39. BISHOP RAYS

Name every square a bishop on g2 attacks on an empty board.

VISUALIZATION DRILLS

Exercises 40-42

40. ROOK RAYS

Name every square a rook on g2 attacks on an empty board.

41. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 Nc6 3. Bc4 Bc5 4. c3 Nf6 5. d4 exd4

42. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 Nc6 3. Bb5 a6 4. Ba4 Nf6 5. O-O Be7

VISUALIZATION DRILLS

Exercises 43-45

43. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 Nc6 3. d4 exd4 4. Nxd4 Nf6 5. Nxc6 bxc6

44. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 Nc6 3. Nc3 Nf6 4. Bb5 Bb4 5. O-O O-O

45. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nc3 Nf6 3. f4 d5 4. fxe5 Nxe4 5. Nf3 Be7

VISUALIZATION DRILLS

Exercises 46-48

46. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. f4 exf4 3. Nf3 g5 4. h4 g4 5. Ne5 Nf6

47. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nf3 Nxe4 5. d4 d5

48. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e5 2. Nf3 d6 3. d4 Nf6 4. Nc3 Nbd7 5. Bc4 Be7

VISUALIZATION DRILLS

Exercises 49-51

49. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 d5 2. exd5 Qxd5 3. Nc3 Qd8 4. d4 Nf6 5. Nf3 c6

50. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 e6 2. d4 d5 3. Nc3 Nf6 4. e5 Nfd7 5. f4 c5

51. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c6 2. d4 d5 3. e5 Bf5 4. Nf3 e6 5. Be2 c5

VISUALIZATION DRILLS

Exercises 52-54

52. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 Nc6

53. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 a6

54. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 g6

VISUALIZATION DRILLS

Exercises 55-57

55. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 Nc6

56. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e6

57. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 g6 5. Nc3 Bg7

VISUALIZATION DRILLS

Exercises 58-60

58. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 e6 3. d4 cxd4 4. Nxd4 Nc6 5. Nc3 Qc7

59. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 e6 3. d4 cxd4 4. Nxd4 a6 5. Bd3 Qc7

60. POSITION RECONSTRUCTION

Play this line mentally, then place every piece: 1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5

ANSWER SECTION

Visualization solutions

CORRECTION METHOD

Reconstruct your internal answer before looking. Compare square by square. If one piece is misplaced, replay the sequence from the last position you know with certainty.

VISUALIZATION ANSWERS

Answers 1-3

01. SQUARE COLOR

a1 is dark. Neighbors: b1, a2

02. SQUARE COLOR

b2 is dark. Neighbors: b1, a2, c2, b3

03. SQUARE COLOR

c3 is dark. Neighbors: c2, b3, d3, c4

VISUALIZATION ANSWERS

Answers 4-6

04. SQUARE COLOR

d4 is dark. Neighbors: d3, c4, e4, d5

05. SQUARE COLOR

e5 is dark. Neighbors: e4, d5, f5, e6

06. SQUARE COLOR

f6 is dark. Neighbors: f5, e6, g6, f7

VISUALIZATION ANSWERS

Answers 7-9

07. SQUARE COLOR

g7 is dark. Neighbors: g6, f7, h7, g8

08. SQUARE COLOR

h8 is dark. Neighbors: h7, g8

09. SQUARE COLOR

a8 is light. Neighbors: a7, b8

VISUALIZATION ANSWERS

Answers 10-12

10. SQUARE COLOR

h1 is light. Neighbors: g1, h2

11. SQUARE COLOR

c6 is light. Neighbors: c5, b6, d6, c7

12. SQUARE COLOR

f3 is light. Neighbors: f2, e3, g3, f4

VISUALIZATION ANSWERS

Answers 13-15

13. KNIGHT ROUTE

Minimum distance: 6 knight moves. Verify a route on the board after attempting it blind.

14. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

15. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

VISUALIZATION ANSWERS

Answers 16-18

16. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

17. KNIGHT ROUTE

Minimum distance: 6 knight moves. Verify a route on the board after attempting it blind.

18. KNIGHT ROUTE

Minimum distance: 2 knight moves. Verify a route on the board after attempting it blind.

VISUALIZATION ANSWERS

Answers 19-21

19. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

20. KNIGHT ROUTE

Minimum distance: 4 knight moves. Verify a route on the board after attempting it blind.

21. KNIGHT ROUTE

Minimum distance: 4 knight moves. Verify a route on the board after attempting it blind.

VISUALIZATION ANSWERS

Answers 22-24

22. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

23. KNIGHT ROUTE

Minimum distance: 3 knight moves. Verify a route on the board after attempting it blind.

24. KNIGHT ROUTE

Minimum distance: 4 knight moves. Verify a route on the board after attempting it blind.

VISUALIZATION ANSWERS

Answers 25-27

25. BISHOP RAYS

a1, g1, b2, f2, c3, e3, c5, e5, b6, f6, a7, g7, h8

26. ROOK RAYS

d1, d2, d3, a4, b4, c4, e4, f4, g4, h4, d5, d6, d7, d8

27. BISHOP RAYS

b1, h1, c2, g2, d3, f3, d5, f5, c6, g6, b7, h7, a8

VISUALIZATION ANSWERS

Answers 28-30

28. ROOK RAYS

e1, e2, e3, a4, b4, c4, d4, f4, g4, h4, e5, e6, e7, e8

29. BISHOP RAYS

a1, e1, b2, d2, b4, d4, a5, e5, f6, g7, h8

30. ROOK RAYS

c1, c2, a3, b3, d3, e3, f3, g3, h3, c4, c5, c6, c7, c8

VISUALIZATION ANSWERS

Answers 31-33

31. BISHOP RAYS

a1, b2, c3, d4, h4, e5, g5, e7, g7, d8, h8

32. ROOK RAYS

f1, f2, f3, f4, f5, a6, b6, c6, d6, e6, g6, h6, f7, f8

33. BISHOP RAYS

b2, c3, d4, e5, f6, g7, h8

VISUALIZATION ANSWERS

Answers 34-36

34. ROOK RAYS

b1, c1, d1, e1, f1, g1, h1, a2, a3, a4, a5, a6, a7, a8

35. BISHOP RAYS

a1, b2, c3, d4, e5, f6, g7

36. ROOK RAYS

h1, h2, h3, h4, h5, h6, h7, a8, b8, c8, d8, e8, f8, g8

VISUALIZATION ANSWERS

Answers 37-39

37. BISHOP RAYS

f1, e2, d3, a4, c4, a6, c6, d7, e8

38. ROOK RAYS

b1, b2, b3, b4, a5, c5, d5, e5, f5, g5, h5, b6, b7, b8

39. BISHOP RAYS

f1, h1, f3, h3, e4, d5, c6, b7, a8

VISUALIZATION ANSWERS

Answers 40-42

40. ROOK RAYS

g1, a2, b2, c2, d2, e2, f2, h2, g3, g4, g5, g6, g7, g8

41. POSITION RECONSTRUCTION

FEN after the sequence: r1bqk2r/pppp1ppp/2n2n2/2b5/2BpP3/2P2N2/PP3PPP/RNBQK2R w KQkq - 0 6

42. POSITION RECONSTRUCTION

FEN after the sequence: r1bqk2r/1pppbppp/p1n2n2/4p3/B3P3/5N2/PPPP1PPP/RNBQ1RK1 w kq - 4 6

VISUALIZATION ANSWERS

Answers 43-45

43. POSITION RECONSTRUCTION

FEN after the sequence: r1bqkb1r/p1pp1ppp/2p2n2/8/4P3/8/PPP2PPP/RNBQKB1R w KQkq - 0 6

44. POSITION RECONSTRUCTION

FEN after the sequence: r1bq1rk1/pppp1ppp/2n2n2/1B2p3/1b2P3/2N2N2/PPPP1PPP/R1BQ1RK1 w - - 8 6

45. POSITION RECONSTRUCTION

FEN after the sequence: rnbqk2r/ppp1bppp/8/3pP3/4n3/2N2N2/PPPP2PP/R1BQKB1R w KQkq - 2 6

VISUALIZATION ANSWERS

Answers 46-48

46. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/pppp1p1p/5n2/4N3/4PppP/8/PPPP2P1/RNBQKB1R w KQkq - 2 6

47. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/ppp2ppp/8/3p4/3Pn3/5N2/PPP2PPP/RNBQKB1R w KQkq - 0 6

48. POSITION RECONSTRUCTION

FEN after the sequence: r1bqk2r/pppnbppp/3p1n2/4p3/2BPP3/2N2N2/PPP2PPP/R1BQK2R w KQkq - 5 6

VISUALIZATION ANSWERS

Answers 49-51

49. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/pp2pppp/2p2n2/8/3P4/2N2N2/PPP2PPP/R1BQKB1R w KQkq - 0 6

50. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/pp1n1ppp/4p3/2ppP3/3P1P2/2N5/PPP3PP/R1BQKBNR w KQkq - 0 6

51. POSITION RECONSTRUCTION

FEN after the sequence: rn1qkbnr/pp3ppp/4p3/2ppPb2/3P4/5N2/PPP1BPPP/RNBQK2R w KQkq - 0 6

VISUALIZATION ANSWERS

Answers 52-54

52. POSITION RECONSTRUCTION

FEN after the sequence: r1bqkb1r/pp2pppp/2np1n2/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 3 6

53. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/1p2pppp/p2p1n2/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 0 6

54. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/pp2pp1p/3p1np1/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 0 6

VISUALIZATION ANSWERS

Answers 55-57

55. POSITION RECONSTRUCTION

FEN after the sequence: r1bqkb1r/pp2pppp/2np1n2/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 3 6

56. POSITION RECONSTRUCTION

FEN after the sequence: rnbqkb1r/pp3ppp/3ppn2/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 0 6

57. POSITION RECONSTRUCTION

FEN after the sequence: r1bqk1nr/pp1pppbp/2n3p1/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 2 6

VISUALIZATION ANSWERS

Answers 58-60

58. POSITION RECONSTRUCTION

FEN after the sequence: r1b1kbnr/ppqp1ppp/2n1p3/8/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 3 6

59. POSITION RECONSTRUCTION

FEN after the sequence: rnb1kbnr/1pqp1ppp/p3p3/8/3NP3/3B4/PPP2PPP/RNBQK2R w KQkq - 2 6

60. POSITION RECONSTRUCTION

FEN after the sequence: r1bqkb1r/pp1p1ppp/2n2n2/4p3/3NP3/2N5/PPP2PPP/R1BQKB1R w KQkq - 0 6

REVIEW

Four-week board-vision plan

1

Week 1: square colors and coordinates for five minutes daily.

2

Week 2: knight routes and long-range rays.

3

Week 3: reconstruct ten-move opening sequences.

4

Week 4: calculate three positions without moving the pieces.

RETRIEVAL SCHEDULE

Repeat missed positions tomorrow, then after 3, 7, 14, and 30 days. Remove a position only after two correct solutions with a complete explanation.